Introduction to Probability and Statistics

(EE 354 / CE 361)

Elements of A Probabilistic Model

Today's Agenda

- Recall: Basic Set Notation and Theory & De Morgans's Laws
- Consequences of the Axioms of Probability
- A systematic approach to dealing w/ uncertainty
- Probability Law: Example w/ Countably Infinite Sample Space
- Further examples for practice

Discrete Uniform Law

- Let all outcomes be equally likely
- Then,

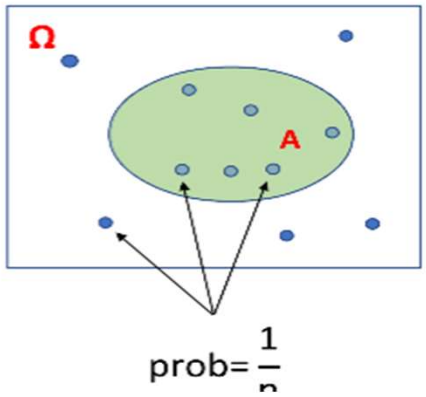
$$\mathbf{P}(A) = \frac{number\ of\ elements\ of\ A}{total\ number\ of\ sample\ points}$$

- Computing probabilities ≡ counting
- Defines fair coins, fair dice, well-shuffled decks

Discrete uniform law

finite

- Assume Ω consists of n equally likely elements
- Assume A consists of k elements



$$P(A) = k * \frac{1}{n}$$

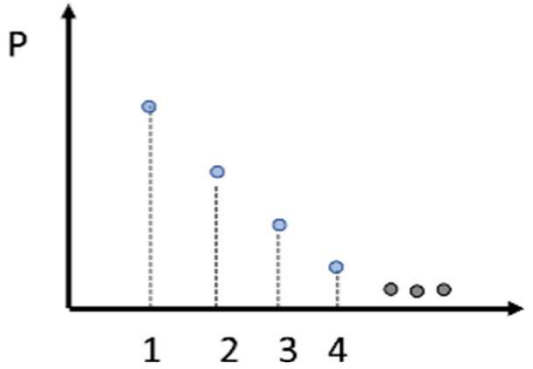
Probability calculation: discrete but infinite sample space

Specify the sample space

1. Sample space: {1, 2, . . . }

Specify the probability law

2. We are given $P(n)=\frac{1}{2^n}$, n=1, 2, . . .



$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{2} * \frac{1}{1-\frac{1}{2}} = 1$$

Identify an event of interest

3. P (outcome is even) = P ({2, 4, 6, . . . })

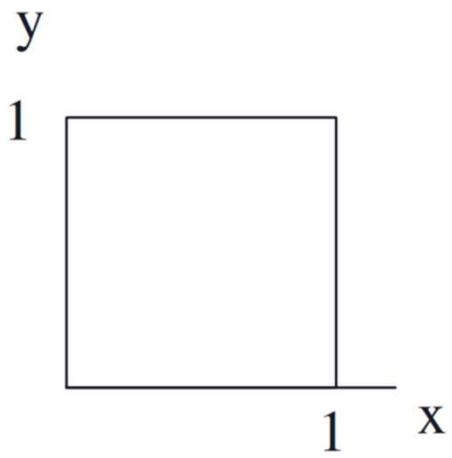
Calculate the probability

$$= P(\{2\} \cup \{4\} \cup \{6\} \cup \ldots) = P(2) + P(4) + P(6) + \ldots$$

$$= \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \ldots = \frac{1}{4}\left(1 + \frac{1}{4} + \frac{1}{4^2} + \ldots\right) = \frac{1}{4} * \frac{1}{1-\frac{1}{4}} = \frac{1}{3}$$

Continuous Uniform Law

- Two "random" numbers in $[0,1]$.



- Uniform law: Probability = Area
  - $\mathbf{P}\left(X+Y \leq \frac{1}{2}\right) =$
  - $\mathbf{P}\big((X,Y) = (0.5, 0.3)\big) =$

Axioms of Probability

1. Nonnegativity: $\mathbf{P}(A) \geq 0$
2. Normalization: $\mathbf{P}(\Omega) = 1$
3. Additivity: If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

Some consequences of the axioms…

Axioms

$P(A) \geq 0$

$P(\Omega) = 1$

For disjoint events:

$P(A \cup B) = P(A) + P(B)$

Consequences

$P(A) \leq 1$

$P(\emptyset) = 0$

$P(A) + P(A^c) = 1$

$P(A \cup B \cup C) = P(A) + P(B) + P(C)$

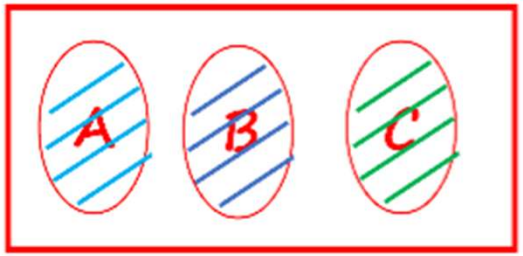
and similarly for *k* disjoint events

$P(\{s_1, s_2, \ldots, s_k\}) = P(\{s_1\}) + \ldots + P(\{s_k\})$

$= P(s_1) + \ldots + P(s_k)$

Some simple consequences of the axioms

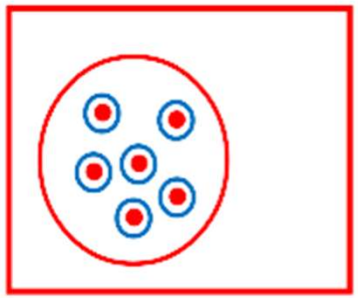
- A, B, C disjoint: P (A U B U C) = P (A) + P (B) + P (C)



$$P(A \cup B \cup C) = P((A \cup B) \cup C) = P(A \cup B) + P(C)$$
$$= P(A) + P(B) + P(C)$$

$$\text{If } A_1, \ldots A_k \text{ disjoint} \Rightarrow P(A_1 \cup \ldots \cup A_k) = \sum_{i=1}^{k} P(A_i)$$

- $P(\{s_1, s_2, \ldots, s_k\}) = P(\{s_1\} \cup \{s_2\} \cup \ldots \cup \{s_k\})$

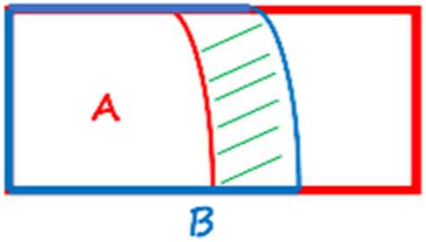


$$= P(\{s_1\} + \{s_2\} + \ldots + \{s_k\})$$

$$= P(s_1) + P(s_2) + \ldots + P(s_k)$$

More consequences of the axioms

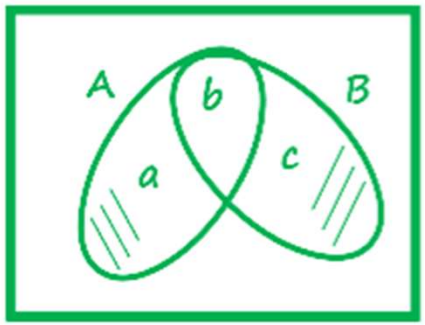
- If $A \subset B$, then $P(A) \leq P(B)$



$B = A \cup (B \cap A^c)$

$P(B) = P(A) + \underbrace{P(B \cap A^c)}_{\geq 0} \geq P(A)$

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



$a = P(A \cap B^c) \quad b = P(A \cap B) \quad c = P(B \cap A^c)$

$P(A \cup B) = a + b + c$

$P(A) + P(B) - P(A \cap B) = (a + b) + (b + c) - b$

$= a + b + c$

- $P(A \cup B) \leq P(A) + P(B)$ union bound

Choosing the Right Sample Space

Example 1.1. Consider two alternative games, both involving ten successive coin tosses:

Game 1: We receive \$1 each time a head comes up.

Game 2: We receive \$1 for every coin toss, up to and including the first time a head comes up. Then, we receive \$2 for every coin toss, up to the second time a head comes up. More generally, the dollar amount per toss is doubled each time a head comes up.

What is an appropriate sample space for each of these experiments?

Probability Law: Explicit vs Implicit/Indirect Specification

- Explicit Specification
  - List out (enumerate) the assigned probability of each subset of sample space

- Implicit Specification
  - For the experiments with finite sample spaces, Discrete Probability Law shows that in probabilities of single-element events are sufficient to characterize (or specify) the probability law.
  - We can use the probabilities of single-element events and Discrete Probability Law to figure out the probability of any other event (i.e the events with more than one element).

Utilizing Probability Theory for Dealing with Uncertainty: Two Stages

1. Understand what your experiment is

2. Construct a probabilistic model
   - Specify the sample space
   - Specify the probability law (implicitly/indirectly)



3. Utilize the probabilistic model to derive the probabilities of certain events of interest
   - Identify an event of interest
   - Calculate its probability



A Simple Example

Tim has a four-sided dice. When he rolls the dice, all the possible outcomes (1, 2, 3, 4) are equally likely.

What is the probability of outcome being even?

1.

2.

3.

4.

5.

A slightly more complicated example…

Tim has a peculiar four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to result of the dice.

What is the probability of outcome being even?

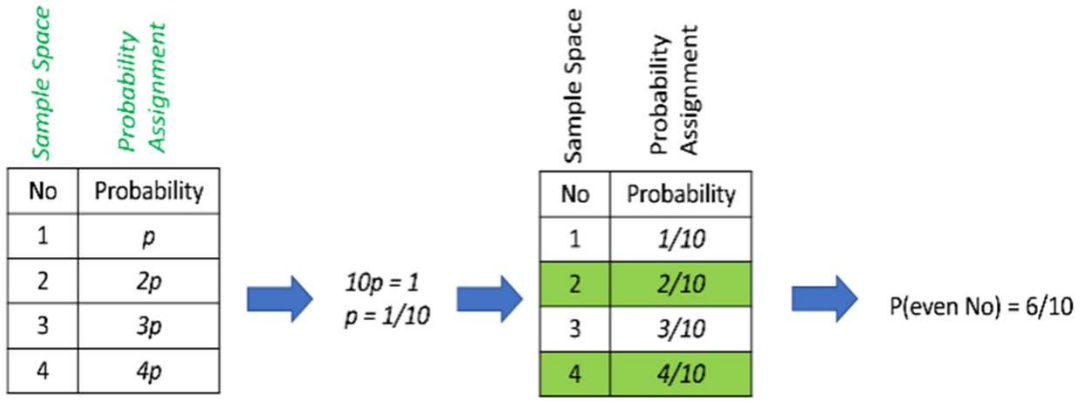
1.

2.

3.

4.

5.



Recall (from Calc 1): the Geometric Series

Given an infinite series of the form:

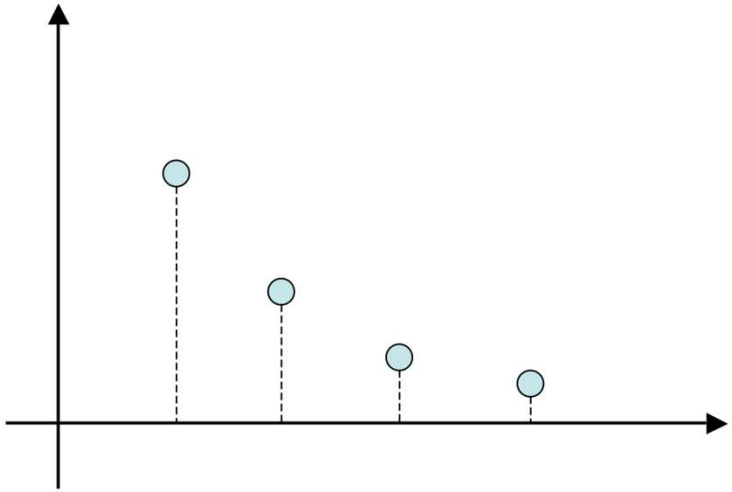
$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \cdots$$

If $|r| < 1$,

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$$

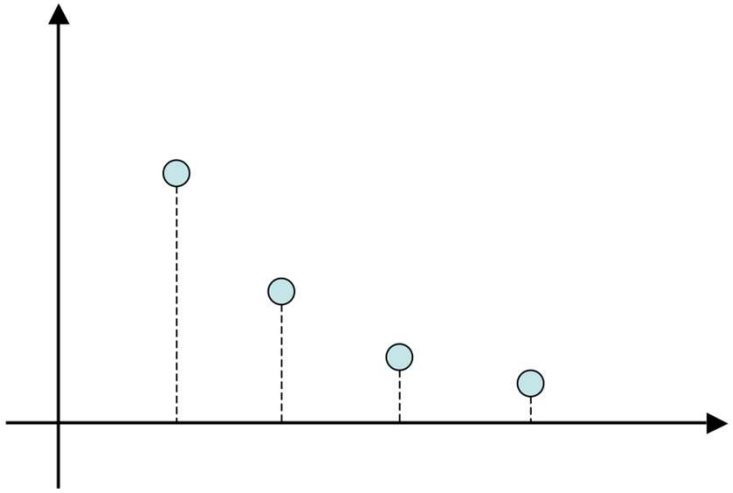
Probability Law: Example w/ Countably Infinite Sample Space

- Sample space: $\{1, 2, 3, \ldots\}$
  - We are given $\mathbf{P}(n) = 2^{-n},\ n = 1, 2, 3, \ldots$
  - Find $\mathbf{P}(\text{outcome is even})$



Probability Law: Example w/ Countably Infinite Sample Space

$$P(\{2,4,6,\dots\}) = P(2) + P(4) + P(6) + \cdots$$
$$= \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \cdots = \frac{1}{3}$$



Corollary of Additivity Axiom

- Countable additivity axiom (needed for this calculation):

  If $A_1, A_2, \dots$ are disjoint events, then:

$$\mathbf{P}(A_1 \cup A_2 \cup \cdots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \cdots$$

Axioms of Probability

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3′. Countable Additivity:

If $A_1, A_2, \ldots$ are disjoint events, then:

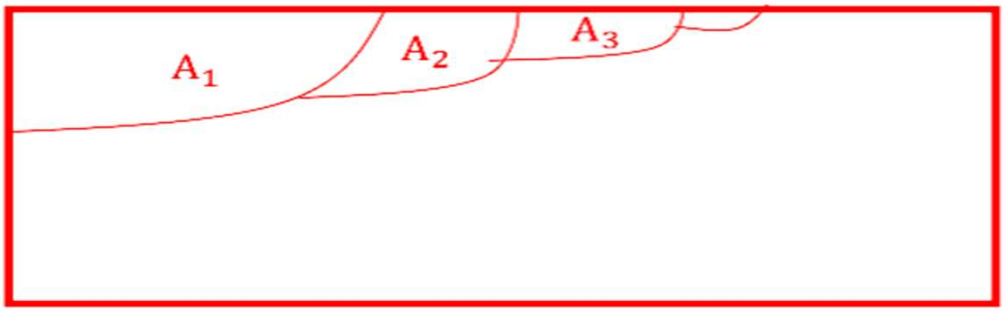
$$\mathbf{P}(A_1 \cup A_2 \cup \cdots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \cdots$$

Countable additivity axiom

- Strengthens the finite additivity axiom

Countable Additivity Axiom:

If $A_1, A_2, A_3, \ldots$ is an infinite sequence of disjoint events,
then $P(A_1 \cup A_2 \cup A_3 \cup \ldots) = P(A_1) + P(A_2) + P(A_3) + \ldots$.





$$1 = P(\Omega) = P\left(\bigcup \{(x, y)\}\right) \overset{?}{=} \sum P(\{(x, y)\}) = \sum 0 = 0$$

- Additivity holds only for **"countable"** sequences of events
- The unit square (similarly, the real line, etc.) is **not countable** (its elements cannot be arranged in a sequence)
- "Area" is a legitimate probability law on the unit square, as long as we do not try to assign probabilities/areas to "very strange" sets

Possibilities for Sample Spaces

1. Discrete and Finite

2. Discrete and Infinite (Countably Infinite)
   - Elements can be arranged in a sequence (i.e. there can be 1-to-1 mapping between elements and integers)
   - Example: Consider repeated coin tosses. After how many coin tosses will you get the first tail.
   - Sample Space = {1, 2, 3, 4, ……..}

3. Continuous (Uncountably Infinite)
   - Elements cannot be arranged in a sequence
   - Example: Throwing a dart on a unit square.

Example 3

Out of the students in a class, 60% love Cricket, 70% love Soccer, and 40% love both Cricket and Soccer. What is the probability that a randomly selected student loves neither Cricket nor Soccer.

1.

2.

3.

4.

5.